$$Zn\,(s) + 2NH_4^+\,(aq) + 2\,MnO_2(s) \rightarrow Zn^{2+}(aq) + Mn_2O_3\,(s) + H_2O\,(l) + 2NH_3 \quad \ldots\ldots (4)$$

Ammonia produced at the cathode combines with Zn^{2+} to form a complex ion $\left[Zn\,(NH_3)_4\right]^{2+}(aq)$. As the reaction proceeds, the concentration of NH_4^+ will decrease and the aqueous NH_3 will increase which lead to the decrease in the emf of cell.

Mercury button cell

Anode	:	zinc amalgamated with mercury
Cathode	:	HgO mixed with graphite
Electrolyte	:	Paste of KOH and ZnO
Oxidation occurs at anode	:	$\overset{0}{Zn}(s) + 2OH^-_{\text{from KOH}}(aq) \rightarrow \overset{+2}{ZnO}(s) + H_2O\,(l) + 2e^-$
Reduction occurs at cathode	:	$\overset{+2}{HgO}(s) + H_2O\,(l) + 2e^- \rightarrow \overset{0}{Hg}(l) + 2OH^-\,(aq)$
Overall reaction	:	$Zn\,(s) + HgO\,(s) \rightarrow ZnO\,(s) + Hg\,(l)$
Cell emf	:	about 1.35V.
Uses	:	It has higher capacity and longer life. Used in pacemakers, electronic watches, cameras etc…

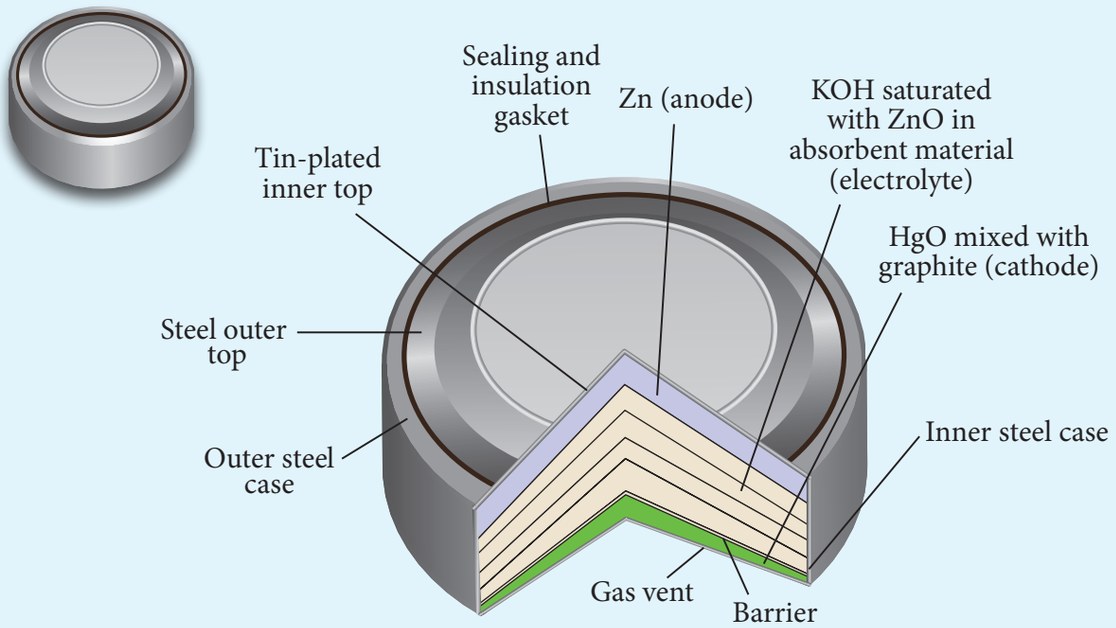


Figure 9.11 Mercury button cell

Secondary batteries

We have already learnt that the electrochemical reactions which take place in a galvanic cell may be reversed by applying a potential slightly greater than the emf generated by the cell. This principle is used in secondary batteries to regenerate the original reactants. Let us understand the function of secondary cell by considering the lead storage battery as an example

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